Chapter 8: Maximum Likelihood Estimation

Chi-Kin Lam

October 19, 2026

CSAI2121 Probability and Statistics
Macao Polytechnic University

Learning Objectives

By the end of this lecture, you should be able to:

- Distinguish between a **parameter**, an **estimator**, and an **estimate**
- Explain the difference between **probability** and **likelihood**
- Construct likelihood and log-likelihood functions for i.i.d. observations
- Find a Maximum Likelihood Estimate using basic differentiation
- Derive the MLE of a Bernoulli probability
- Derive the MLE of the mean and variance of a Gaussian distribution
- Explain why log-likelihood is preferred in practical computation
- Connect MLE to binary cross-entropy, mean squared error, and machine learning

From Data to Unknown Parameters

Motivation: Reversing the Probability Question

In earlier chapters, we usually assumed that the probability model was known.

Forward Probability Problem

Given a parameter value, what data are likely to be observed?

$$\theta \quad \longrightarrow \quad p(\text{data} \mid \theta).$$

In statistical inference, the parameter is usually unknown.

Inverse Statistical Problem

Given the observed data, which parameter value best explains them?

$$\text{observed data} \quad \longrightarrow \quad \text{estimate of } \theta.$$

Maximum Likelihood Estimation provides a systematic answer to this inverse problem.

Statistical Models and Parameters

A **statistical model** is a family of probability distributions indexed by one or more parameters.

Model	Parameter(s)	Interpretation
Bernoulli	p	Probability of success
Poisson	λ	Average event rate
Gaussian	μ, σ^2	Centre and spread
Linear regression	$\mathbf{w}, \sigma^2$	Weights and noise variance

General Notation

We write:

$$X \sim p(x \mid \theta),$$

where θ represents the unknown parameter or collection of parameters.

The data are observed, while θ is unknown and must be estimated.

Parameter, Estimator, and Estimate

- A **parameter** is a fixed but unknown feature of the population:

$$\theta.$$

- An **estimator** is a rule or function of random observations:

$$\widehat{\theta} = g(X_1, \ldots, X_n).$$

- An **estimate** is the numerical value obtained after observing data:

$$\widehat{\theta}_{\text{observed}} = g(x_1, \ldots, x_n).$$

Example

The population mean μ is a parameter. The sample mean

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

is an estimator. If the observed sample mean is 6.4, then 6.4 is the estimate.

Probability vs. Likelihood

The same mathematical expression can be viewed in two different ways.

	Probability	**Likelihood**
Fixed quantity	Parameter θ	Observed data x
Variable quantity	Possible data x	Possible parameter θ
Main question	What data may occur?	Which parameter best explains the observed data?
Notation	$p(x \mid \theta)$	$L(\theta; x)$

Key Point

A likelihood is viewed as a function of the unknown parameter, with the observed data held fixed.

A Simple Coin Example

Suppose a coin has an unknown probability p of landing heads.

We flip it three times and observe:

$$H,\ H,\ T.$$

Under the Bernoulli model:

$$P(H, H, T \mid p) = p \cdot p \cdot (1-p) = p^2(1-p).$$

Once the data H, H, T have been observed, define:

$$L(p) = p^2(1-p).$$

- If $p = 0.1$, the observed sequence has likelihood $0.1^2(0.9) = 0.009$.
- If $p = 0.5$, the likelihood is $0.5^2(0.5) = 0.125$.
- If $p = 2/3$, the likelihood is $(2/3)^2(1/3) \approx 0.148$.

The value $p = 2/3$ explains this particular dataset better than $p = 0.1$ or $p = 0.5$.

The Likelihood Function

Definition

For observed data $x_1, \ldots, x_n$ from a model with parameter θ, the likelihood function is

$$L(\theta; x_1, \ldots, x_n) = p(x_1, \ldots, x_n \mid \theta).$$

If the observations are independent:

$$p(x_1, \ldots, x_n \mid \theta) = \prod_{i=1}^{n} p(x_i \mid \theta).$$

Therefore:

$$\boxed{L(\theta) = \prod_{i=1}^{n} p(x_i \mid \theta)}.$$

Role of Independence

Independence allows the joint probability or density to be written as a product of individual terms.

Why Does the Likelihood Use a Product?

Suppose $X_1, \ldots, X_n$ are independent observations.

For two observations:

$$p(x_1, x_2 \mid \theta) = p(x_1 \mid \theta) p(x_2 \mid \theta).$$

For n observations:

$$p(x_1, \ldots, x_n \mid \theta) = \prod_{i=1}^{n} p(x_i \mid \theta).$$

- A parameter receives high likelihood only if it gives reasonably high probability or density to **all** observed data points.
- A single observation that is extremely unlikely under the model can greatly reduce the product.
- The product combines evidence from the whole sample.

Important Assumption

If the observations are dependent, simply multiplying the individual probabilities may be incorrect.

Likelihood Is Not a Posterior Probability

A common mistake is to interpret:

$$L(\theta) = p(\text{data} \mid \theta)$$

as the probability that θ is correct.

This is not generally valid.

- Likelihood:
$$p(\text{data} \mid \theta).$$
- Posterior probability:
$$p(\theta \mid \text{data}).$$

Bayes' theorem connects them:

$$p(\theta \mid \text{data}) = \frac{p(\text{data} \mid \theta)p(\theta)}{p(\text{data})}.$$

Key Distinction

MLE uses only the likelihood. MAP estimation, introduced later, also includes a prior distribution $p(\theta)$.

Maximum Likelihood and Log-Likelihood

Maximum Likelihood Estimation

Definition

The Maximum Likelihood Estimate is the parameter value that maximizes the likelihood of the observed data:

$$\boxed{\widehat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} L(\theta)}.$$

- $\max_\theta L(\theta)$ is the maximum likelihood **value**.
- $\arg\max_\theta L(\theta)$ is the parameter value at which the maximum occurs.

Plain-Language Meaning

Choose the parameter value under which the observed dataset would be most plausible.

The Log-Likelihood

Products of many probabilities can be difficult to manipulate.

Define the **log-likelihood**:

$$\ell(\theta) = \log L(\theta).$$

Since:

$$L(\theta) = \prod_{i=1}^{n} p(x_i \mid \theta),$$

we obtain:

$$\boxed{\ell(\theta) = \sum_{i=1}^{n} \log p(x_i \mid \theta)}.$$

Because the logarithm is strictly increasing:

$$\arg\max_{\theta} L(\theta) = \arg\max_{\theta} \ell(\theta).$$

Maximizing likelihood and maximizing log-likelihood give the same parameter estimate.

Why Use the Log-Likelihood?

1. **Products become sums:**

$$\log\left(\prod_{i=1}^{n} a_i\right) = \sum_{i=1}^{n} \log a_i.$$

2. **Differentiation becomes easier.**
3. **Improved numerical stability:**
   Multiplying many small probabilities can produce extremely small numbers. For example:

$$0.01^{100} = 10^{-200}.$$

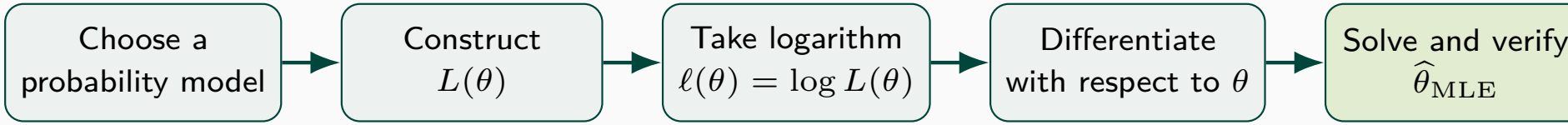
   Instead:

$$\log(0.01^{100}) = 100\log(0.01) \approx -460.5.$$

Practical Rule

In statistical software and machine learning, we normally maximize log-likelihood or minimize negative log-likelihood.

General MLE Workflow



1. Specify the distribution and its parameter.
2. Insert the observed data into the likelihood.
3. Simplify using the log-likelihood.
4. Find candidate maximizing values.
5. Check that the candidate is valid and gives a maximum.

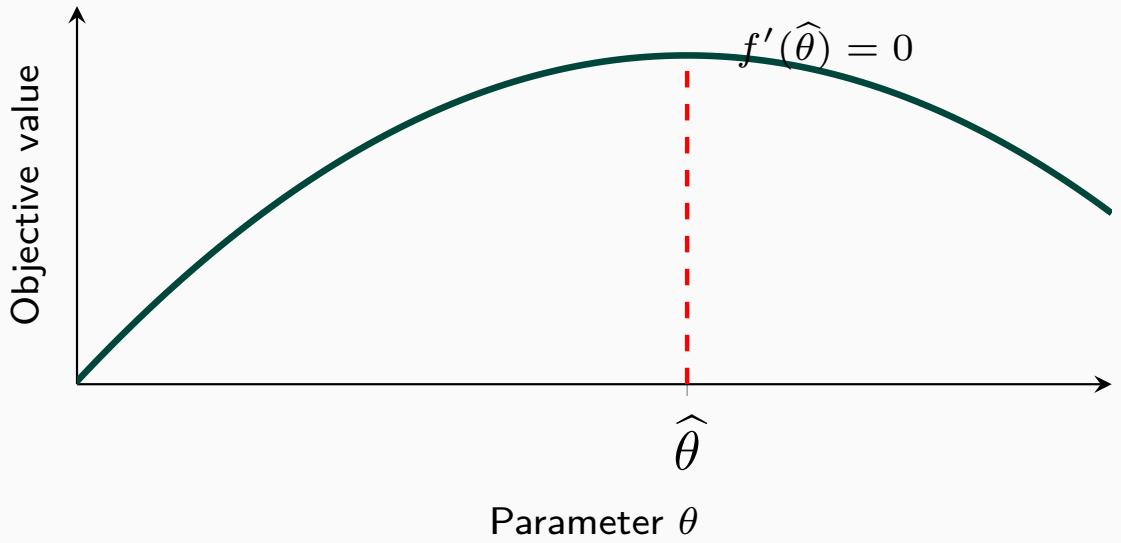
Calculus Refresher: Finding a Maximum

For a differentiable function $f(\theta)$, an interior maximum often satisfies:

$$f'(\theta) = 0.$$

To verify a maximum, we may use:

$$f''(\theta) < 0.$$



Restriction

The parameter must remain in its valid range, such as $0 \leq p \leq 1$ or $\sigma^2 > 0$.

Interior and Boundary Solutions

Solving: $\ell'(\theta) = 0$ only finds possible **interior** optima.

We must also consider:

- the valid parameter space;
- endpoints or boundaries;
- whether the likelihood is increasing or decreasing;
- whether the candidate is a maximum rather than a minimum.

Example: Bernoulli Parameter

The valid range is:

$$0 \leq p \leq 1.$$

If all observations are failures, the maximum occurs at the boundary $p = 0$. If all are successes, it occurs at $p = 1$.

Practical Message

Differentiation is part of MLE, but checking the parameter constraints is equally important.

Bernoulli Maximum Likelihood Estimation

Bernoulli Model

Suppose:

$$X_i \sim \text{Bernoulli}(p), \qquad X_i \in \{0, 1\}.$$

The PMF is:

$$P(X_i = x_i \mid p) = p^{x_i}(1-p)^{1-x_i}.$$

This compact formula gives:

$$x_i = 1 \quad \Rightarrow \quad P(X_i = 1) = p,$$

$$x_i = 0 \quad \Rightarrow \quad P(X_i = 0) = 1 - p.$$

- p is unknown.
- We observe $x_1, \ldots, x_n$.
- Our goal is to estimate p.

Bernoulli Likelihood

For independent observations:

$$L(p) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i}$$
$$= p^{\sum_{i=1}^{n} x_i}(1-p)^{n-\sum_{i=1}^{n} x_i}.$$

Let:

$$k = \sum_{i=1}^{n} x_i$$

be the number of successes.

Then:

$$\boxed{L(p) = p^k(1-p)^{n-k}}.$$

Important Observation

For estimating p, the order of the observations is irrelevant. The likelihood depends only on the number of successes k.

Bernoulli Log-Likelihood

Take logarithms:

$$\ell(p) = \log L(p).$$

Therefore:

$$\begin{aligned}\ell(p) &= \log\left[p^k(1-p)^{n-k}\right] \\ &= k\log p + (n-k)\log(1-p).\end{aligned}$$

Equivalently:

$$\boxed{\ell(p) = \sum_{i=1}^{n}\left[x_i \log p + (1-x_i)\log(1-p)\right]}.$$

The product in the likelihood has become a sum in the log-likelihood.

Differentiating the Bernoulli Log-Likelihood

For $0 < p < 1$:

$$\ell(p) = k \log p + (n - k) \log(1 - p).$$

Differentiate:

$$\ell'(p) = \frac{k}{p} - \frac{n - k}{1 - p}.$$

Set the derivative equal to zero:

$$\frac{k}{p} - \frac{n - k}{1 - p} = 0.$$

Multiply by $p(1 - p)$:

$$k(1 - p) - (n - k)p = 0.$$

Simplifying:

$$k - np = 0.$$

Bernoulli MLE Result

Solving:

$$k - np = 0$$

gives:

$$\widehat{p}_{\mathrm{MLE}} = \frac{k}{n}.$$

Since $k = \sum_{i=1}^{n} x_i$:

$$\boxed{\widehat{p}_{\mathrm{MLE}} = \frac{1}{n}\sum_{i=1}^{n} x_i = \overline{x}}.$$

The second derivative is:

$$\ell''(p) = -\frac{k}{p^2} - \frac{n-k}{(1-p)^2} < 0$$

for an interior solution.

Interpretation

The MLE of a Bernoulli success probability is simply the observed proportion of successes.

Worked Example: Estimating a Click Probability

Suppose ten users are shown an advertisement.

The observed results are:

$$1,\ 0,\ 1,\ 0,\ 0,\ 1,\ 0,\ 1,\ 0,\ 0,$$

where 1 means "clicked".

The number of clicks is:

$$k = 4, \qquad n = 10.$$

Therefore:

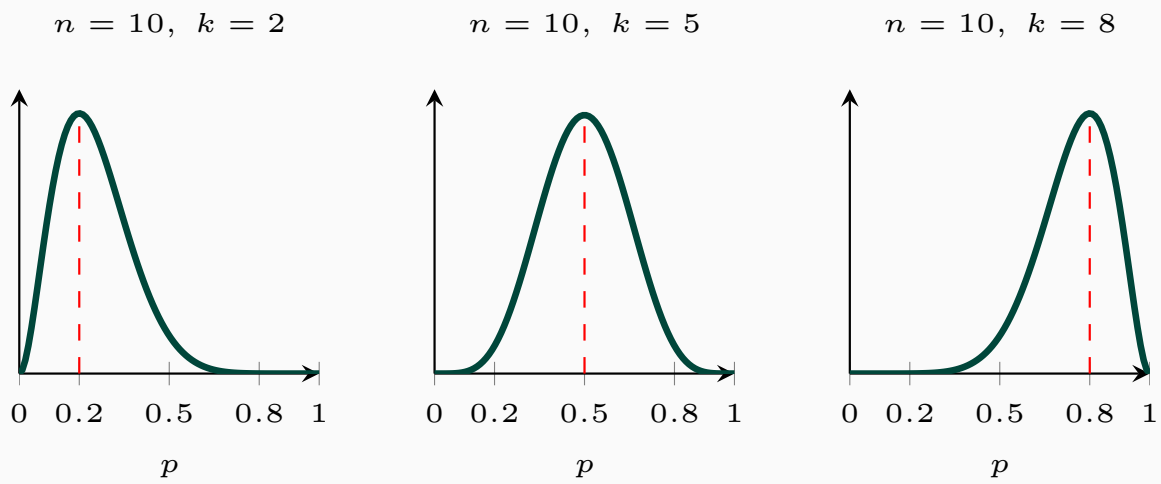
$$\widehat{p}_{\mathrm{MLE}} = \frac{k}{n} = \frac{4}{10} = 0.4.$$

Interpretation

Based on this sample, the parameter value that maximizes the likelihood is an estimated click probability of 40%.

This does not mean the true population click probability must equal exactly 0.4.

Visualizing Bernoulli Likelihoods



- The likelihood peak occurs at $p = k/n$.
- More observed successes move the peak to the right.
- The red dashed line marks the MLE.

Boundary Cases

The differentiation assumed: $0 < p < 1$. However, special cases can occur.

- If $k = 0$, all observations are failures:

$$L(p) = (1 - p)^n.$$

  The likelihood is maximized at:

$$\widehat{p}_{\text{MLE}} = 0.$$

- If $k = n$, all observations are successes:

$$L(p) = p^n.$$

  The likelihood is maximized at:

$$\widehat{p}_{\text{MLE}} = 1.$$

Caution

With a small sample, an MLE of 0 or 1 may be too extreme for reliable prediction. This motivates smoothing, regularization, and Bayesian methods.

AI Connection: Binary Cross-Entropy

For Bernoulli observations:

$$\ell(p) = \sum_{i=1}^{n} \left[x_i \log p + (1 - x_i) \log(1 - p) \right].$$

The negative log-likelihood is:

$$-\ell(p) = -\sum_{i=1}^{n} \left[x_i \log p + (1 - x_i) \log(1 - p) \right].$$

If a classifier produces a separate predicted probability p_i for each observation:

$$\boxed{\mathcal{L}_{\text{BCE}} = -\sum_{i=1}^{n} \left[x_i \log p_i + (1 - x_i) \log(1 - p_i) \right].}$$

Key Connection

Binary cross-entropy is the negative log-likelihood of a Bernoulli classification model.

Gaussian Mean Estimation

Gaussian Model with Known Variance

Suppose:

$$X_i \sim N(\mu, \sigma^2),$$

where:

- the mean μ is unknown;
- the variance σ^2 is known;
- $X_1, \ldots, X_n$ are independent.

The density of one observation is:

$$p(x_i \mid \mu) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right).$$

Our goal is to estimate the unknown centre μ from the observed data.

Gaussian Likelihood and Log-Likelihood

The likelihood is:

$$L(\mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right).$$

Combining the terms:

$$L(\mu) = \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^n \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\right).$$

Taking logarithms:

$$\boxed{\ell(\mu) = -\frac{n}{2}\log(2\pi) - n\log\sigma - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2}.$$

Maximizing Likelihood Means Minimizing Squared Error

In:

$$\ell(\mu) = -\frac{n}{2}\log(2\pi) - n\log\sigma - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2,$$

the first two terms do not depend on μ.

Therefore:

$$\arg\max_{\mu} \ell(\mu) = \arg\min_{\mu} \sum_{i=1}^{n}(x_i - \mu)^2.$$

Key Connection

Under a Gaussian noise assumption, maximizing likelihood is equivalent to minimizing the sum of squared errors.

This result will later explain why least-squares regression is connected to Gaussian MLE.

Deriving the Gaussian Mean MLE

Differentiate the log-likelihood with respect to μ:

$$\frac{d\ell}{d\mu} = \frac{1}{\sigma^2} \sum_{i=1}^{n} (x_i - \mu).$$

Set the derivative equal to zero:

$$\sum_{i=1}^{n} (x_i - \mu) = 0.$$

Solving for μ:

$$\boxed{\widehat{\mu}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^{n} x_i = \overline{x}}.$$

Also:

$$\frac{d^2\ell}{d\mu^2} = -\frac{n}{\sigma^2} < 0,$$

confirming a maximum.

Worked Example: Gaussian Mean

Suppose the observed data are:

$$4,\ 5,\ 6,\ 7,\ 8.$$

The MLE is:

$$\widehat{\mu}_{\mathrm{MLE}} = \frac{4+5+6+7+8}{5} = 6.$$

Compare the sums of squared errors:

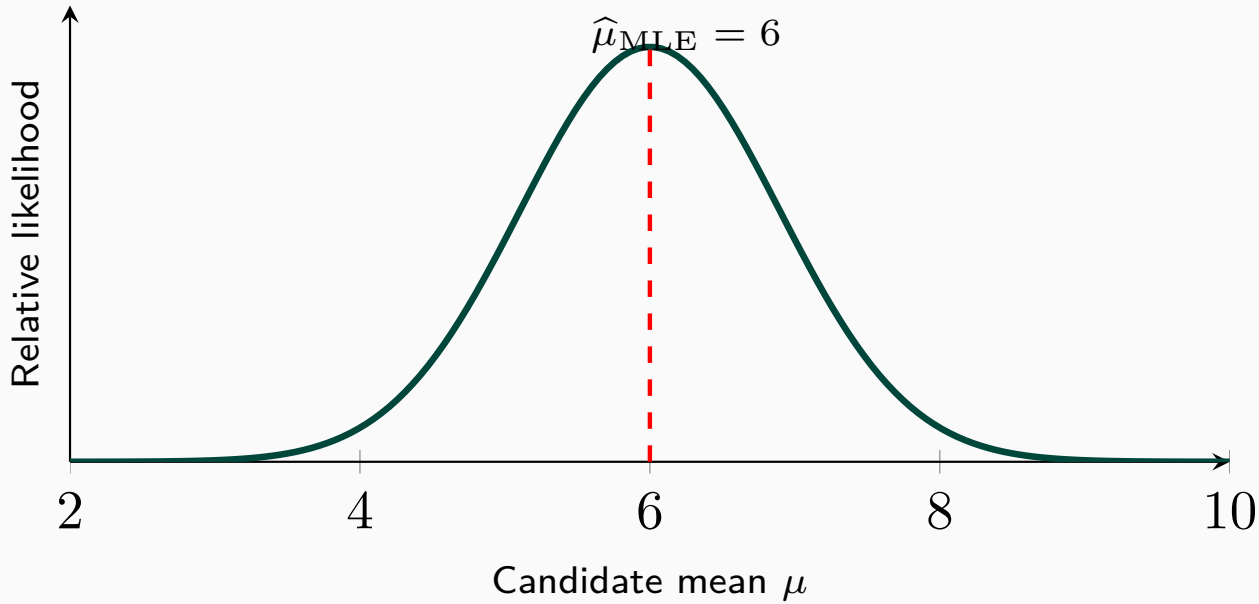
Candidate μ	$\sum_{i=1}^{n}(x_i - \mu)^2$
5	15
6	10
7	15

Interpretation

The sample mean $\mu = 6$ produces the smallest squared error and therefore the largest Gaussian likelihood.

Visualizing the Gaussian Mean Likelihood

For the observations $4, 5, 6, 7, 8$ and known $\sigma = 2$, the relative likelihood has its peak at $\mu = 6$.



Candidate means far from the centre of the observations produce lower likelihood.

AI Connection: Gaussian MLE and Mean Squared Error

Suppose a model predicts:

$$\widehat{y}_i = f(\mathbf{x}_i; \mathbf{w}),$$

and the observed target satisfies:

$$y_i = f(\mathbf{x}_i; \mathbf{w}) + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2).$$

The Gaussian negative log-likelihood contains:

$$\sum_{i=1}^{n} \left(y_i - f(\mathbf{x}_i; \mathbf{w})\right)^2 .$$

Therefore:

$$\boxed{\text{Gaussian MLE} \iff \text{minimizing squared prediction error}}.$$

Machine-Learning Interpretation

Mean Squared Error is not an arbitrary loss: it corresponds to a Gaussian observation or noise model.

Estimating Gaussian Mean and Variance

A Minimal Introduction to Partial Derivatives

Suppose a function depends on two parameters:

$$f(\mu, v).$$

A partial derivative changes one parameter while temporarily treating the other as constant:

$$\frac{\partial f}{\partial \mu} \quad \text{means differentiate with respect to } \mu, \text{ holding } v \text{ fixed,}$$

$$\frac{\partial f}{\partial v} \quad \text{means differentiate with respect to } v, \text{ holding } \mu \text{ fixed.}$$

For a possible interior maximum, we solve:

$$\frac{\partial f}{\partial \mu} = 0, \qquad \frac{\partial f}{\partial v} = 0.$$

Scope of This Course

We only use partial derivatives as repeated single-variable differentiation. No advanced multivariable calculus is required.

Gaussian Model with Two Unknown Parameters

Suppose:

$$X_i \sim N(\mu, \sigma^2),$$

where both μ and σ^2 are unknown.

To simplify the differentiation, write:

$$v = \sigma^2, \qquad v > 0.$$

The log-likelihood is:

$$\boxed{\ell(\mu, v) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log v - \frac{1}{2v}\sum_{i=1}^{n}(x_i - \mu)^2}.$$

We estimate the two parameters by solving:

$$\frac{\partial \ell}{\partial \mu} = 0, \qquad \frac{\partial \ell}{\partial v} = 0.$$

Estimating the Gaussian Mean

Holding v fixed:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{v} \sum_{i=1}^{n} (x_i - \mu).$$

Set equal to zero:

$$\sum_{i=1}^{n} (x_i - \mu) = 0.$$

Therefore:

$$\boxed{\widehat{\mu}_{\mathrm{MLE}} = \overline{x}}.$$

- The estimate of μ remains the sample mean.
- This result does not change when the variance is also unknown.

Estimating the Gaussian Variance

Holding μ fixed:

$$\frac{\partial \ell}{\partial v} = -\frac{n}{2v} + \frac{1}{2v^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Set equal to zero:

$$-\frac{n}{2v} + \frac{1}{2v^2}\sum_{i=1}^{n}(x_i - \mu)^2 = 0.$$

Multiply by $2v^2$:

$$-nv + \sum_{i=1}^{n}(x_i - \mu)^2 = 0.$$

Hence:

$$v = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Gaussian Variance MLE

Substitute:

$$\widehat{\mu}_{\mathrm{MLE}} = \overline{x}.$$

Then:

$$\boxed{\widehat{\sigma}^2_{\mathrm{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2}.$$

Therefore, the Gaussian MLEs are:

$$\boxed{\widehat{\mu}_{\mathrm{MLE}} = \overline{x}, \qquad \widehat{\sigma}^2_{\mathrm{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2}.$$

Interpretation

The fitted Gaussian is centred at the sample mean, with spread determined by the average squared distance from that mean.

Why Does the MLE Variance Use n?

The Gaussian MLE is:

$$\widehat{\sigma}^2_{\mathrm{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2.$$

The usual unbiased sample variance is:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2.$$

They answer slightly different questions:

- Dividing by n maximizes the Gaussian likelihood.
- Dividing by $n-1$ corrects the downward bias:

$$E[S^2] = \sigma^2.$$

Key Point

An MLE is not automatically unbiased. Maximum likelihood and unbiased estimation are different design criteria.

Worked Example: Gaussian Mean and Variance

Suppose:

$$x_1, \ldots, x_5 = 3,\ 5,\ 4,\ 6,\ 2.$$

First:

$$\widehat{\mu}_{\mathrm{MLE}} = \overline{x} = \frac{3+5+4+6+2}{5} = 4.$$

The squared deviations are:

$$(3-4)^2 + (5-4)^2 + (4-4)^2 + (6-4)^2 + (2-4)^2 = 1+1+0+4+4 = 10.$$

Therefore:

$$\widehat{\sigma}^2_{\mathrm{MLE}} = \frac{10}{5} = 2.$$

The unbiased sample variance would be:

$$S^2 = \frac{10}{4} = 2.5.$$

Properties, Limitations, and AI Applications

Why Is MLE Widely Used?

- **General**: applicable to many probability models.
- **Interpretable**: chooses parameters that best explain the observed data.
- **Data-driven**: does not require a prior distribution.
- **Compatible with optimization**: often leads to standard loss functions.
- **Large-sample performance**: under suitable conditions, MLEs often become close to the true parameter as the sample size increases.
- **Foundation for AI/ML**: many model-training objectives are negative log-likelihoods.

Connection to the LLN

As more observations are collected, the average log-likelihood tends to reflect the underlying data-generating distribution more accurately.

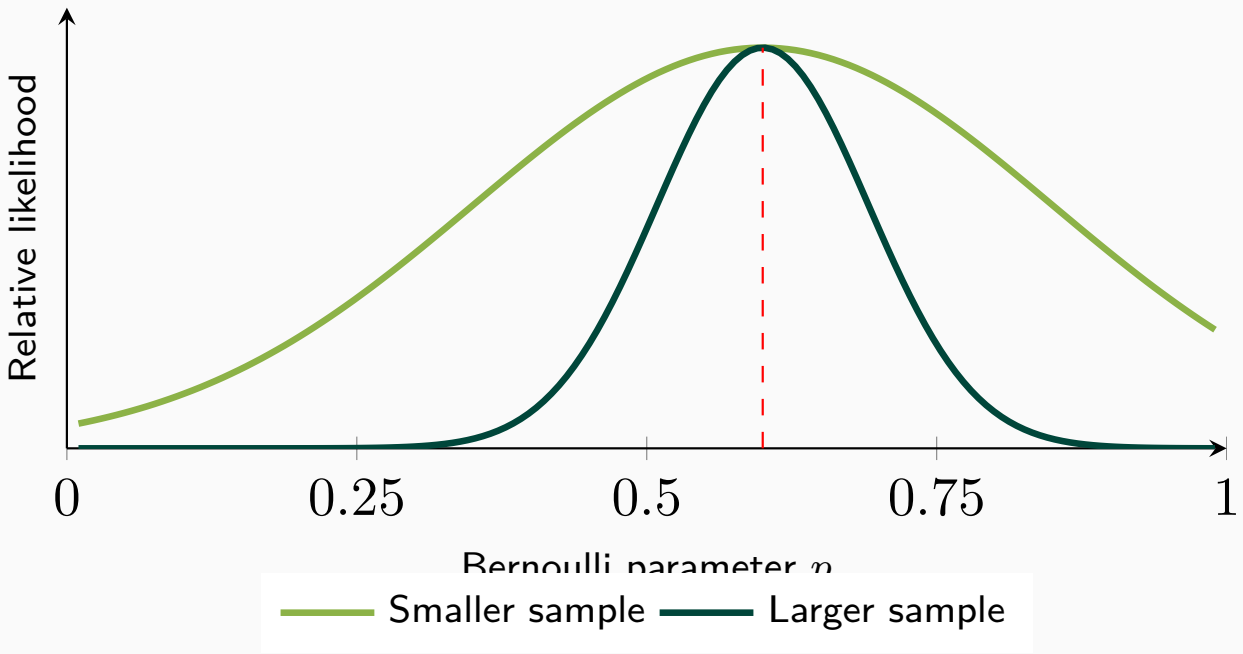
Limitations and Cautions

- **Model dependence**: an MLE can be misleading if the chosen probability model is inappropriate.
- **Small samples**: estimates may be unstable or extreme.
- **Bias**: an MLE need not be unbiased.
- **Multiple maxima**: complicated likelihoods may contain several local maxima.
- **Boundary estimates**: parameters may be estimated at the edge of the valid range.
- **Overfitting**: highly flexible models may fit observed data too closely.

Important

MLE finds the best parameter *within the assumed model*. It does not prove that the model itself is correct.

Effect of Sample Size on the Likelihood



- A larger sample usually produces a more concentrated likelihood.
- A sharper peak indicates that fewer parameter values fit the data well.
- This is related to reduced estimation uncertainty.

AI Applications of Maximum Likelihood

AI/ML task	Probability model	Resulting loss
Binary classification	Bernoulli	Binary cross-entropy
Multi-class classification	Categorical	Categorical cross-entropy
Regression	Gaussian noise	Mean squared error
Count prediction	Poisson	Poisson negative log-likelihood
Density estimation	Gaussian mixture	Mixture negative log-likelihood

Big Picture

Choosing a loss function often corresponds to choosing a probability model and then minimizing its negative log-likelihood.

Summary: Key Definitions and Formulas

$$L(\theta) = \prod_{i=1}^{n} p(x_i \mid \theta)$$

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log p(x_i \mid \theta)$$

$$\widehat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} L(\theta) = \arg\max_{\theta} \ell(\theta)$$

Bernoulli: $\widehat{p}_{\mathrm{MLE}} = \dfrac{1}{n} \sum_{i=1}^{n} x_i$

Gaussian: $\widehat{\mu}_{\mathrm{MLE}} = \overline{x}, \qquad \widehat{\sigma}^2_{\mathrm{MLE}} = \dfrac{1}{n} \sum_{i=1}^{n} (x_i - \overline{x})^2$

Common Mistakes (1/2)

- Treating the likelihood as a probability distribution over θ
- Confusing:

$$p(\text{data} \mid \theta) \quad \text{with} \quad p(\theta \mid \text{data})$$

- Forgetting to multiply the probabilities or densities of independent observations
- Maximizing each observation's probability separately rather than maximizing the joint likelihood
- Forgetting that the observed data are held fixed when viewing likelihood as a function of θ

Common Mistakes (2/2)

- Multiplying many small probabilities instead of using the log-likelihood
- Forgetting parameter restrictions such as:

$$0 \leq p \leq 1, \qquad \sigma^2 > 0$$

- Assuming every solution of $\ell'(\theta) = 0$ must be a maximum
- Forgetting to check boundary cases
- Dividing by $n - 1$ when specifically asked for the Gaussian MLE of variance
- Assuming an MLE must be unbiased
- Assuming that a high maximum likelihood proves the model is correct

Class Exercise

Problem 1: Bernoulli MLE

The observed binary data are:

$$1,\ 0,\ 1,\ 1,\ 0,\ 1,\ 0,\ 1.$$

1. Write the likelihood function $L(p)$.
2. Write the log-likelihood $\ell(p)$.
3. Derive and calculate $\widehat{p}_{\mathrm{MLE}}$.

Problem 2: Gaussian MLE

For the data:

$$3,\ 5,\ 4,\ 6,\ 2,$$

calculate:

1. $\widehat{\mu}_{\mathrm{MLE}}$;
2. $\widehat{\sigma}^2_{\mathrm{MLE}}$;
3. the unbiased sample variance S^2.

Class Exercise: Solution to Problem 1

The data contain: $n = 8, k = 5$ successes.

The likelihood is:

$$L(p) = p^5(1-p)^3.$$

The log-likelihood is:

$$\ell(p) = 5\log p + 3\log(1-p).$$

Differentiate:

$$\ell'(p) = \frac{5}{p} - \frac{3}{1-p}.$$

Setting $\ell'(p) = 0$ gives:

$$5(1-p) - 3p = 0 \quad \Rightarrow \quad 5 - 8p = 0.$$

Therefore:

$$\boxed{\widehat{p}_{\text{MLE}} = \frac{5}{8} = 0.625}.$$

Class Exercise: Solution to Problem 2

The observations are:

$$3,\ 5,\ 4,\ 6,\ 2.$$

The mean MLE is:

$$\widehat{\mu}_{\mathrm{MLE}} = \frac{3+5+4+6+2}{5} = 4.$$

The sum of squared deviations is:

$$(3-4)^2 + (5-4)^2 + (4-4)^2 + (6-4)^2 + (2-4)^2 = 10.$$

Therefore:

$$\boxed{\widehat{\sigma}^2_{\mathrm{MLE}} = \frac{10}{5} = 2}.$$

The unbiased sample variance is:

$$\boxed{S^2 = \frac{10}{5-1} = 2.5}.$$

Connections Ahead

- Gaussian MLE and squared error

  $\longrightarrow$ **Ordinary Least Squares Regression**

- Bernoulli likelihood

  $\longrightarrow$ **Binary Cross-Entropy and Classification**

- Likelihood combined with a prior

  $\longrightarrow$ **Maximum A Posteriori Estimation**

- Gaussian prior on model weights

  $\longrightarrow$ **L2 Regularization**

- Laplace prior on model weights

  $\longrightarrow$ **L1 Regularization**

Preview of Next Week: Linear Regression and MAP

- Today, we estimated the mean of a Gaussian distribution by maximizing likelihood.
- Next week, the Gaussian mean will depend on input features:

$$\mu_i = \mathbf{x}_i^T \mathbf{w}.$$

- We will study the linear regression model:

$$\mathbf{y} = X\mathbf{w} + \boldsymbol{\varepsilon}.$$

- We will connect:

$$\text{Gaussian MLE} \quad \Longleftrightarrow \quad \text{Ordinary Least Squares.}$$

- We will then introduce MAP estimation and explain how prior distributions lead to regularization.

Main Connection

Maximum likelihood provides the probabilistic foundation for fitting many statistical and machine-learning models.